

Solution Architecture Specification

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Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Serves the user interface, manages questionnaire form inputs, and renders the live split-screen generated document workspace (Lead: Deepan).	React.js, Tailwind CSS, TypeScript
API Gateway	Handles centralized routing, input sanitization, rate limiting, and transfers payload distributions securely across backend workers.	FastAPI Routing, AWS API Gateway
Microservices / Core Logic	Orchestrates prompt engineering templates, contextual data lookups, and triggers context evaluations for contract outputs (Leads: M Mohammed Abdul Rahman / Lingaraj S).	Python, LangChain, OpenAI API
External API Integration	Handles secure, cryptographic multi-party signature processing and legally valid electronic sign-offs once documents are finalized (Lead: Sakthi).	DocuSign REST API, SendGrid API
Database Layer	Safely stores account metadata, user system roles, generated document parameters, configuration states, and document signature verification audit logs.	PostgreSQL Database, Redis Cache

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.